

Portfolio Loss Modeling on 2.26 Million Consumer Loans

A written summary of the analysis in this repository. No code required.

Why this study is shaped the way it is

Take a portfolio of correlated risks. Model the distribution of loss it can produce. Reason about the tail, and decide what to charge and what to hold against it.

That description fits a consumer loan book and a property catastrophe treaty equally well. The vocabulary is what diverges: a credit team's **vintage curve** is a reinsurance team's **loss development triangle**; **expected loss** on a book is an ILS investor's **AAL**; a **loss distribution** becomes an **EP curve** when you plot it upside down. Underneath, one set of machinery.

This study runs that machinery end to end on real loan-level data — 2.26 million LendingClub loans originated between 2007 and 2018 — and is framed around portfolio loss throughout rather than around classification accuracy. It answers four questions in order: **when** does loss emerge, **where** does it sit, **what** predicts it, and **how bad can the whole book get**.

1. The data, and the one screen that matters

LendingClub published loan-level performance for every loan it originated: 2,260,701 records, 151 columns, with performance observed through March 2019. The target is binary — did the lender write the loan off, or did the borrower repay in full.

Getting to an honest modeling table means discarding most of the file.

Loans still paying were removed. A loan that is "Current" has no outcome yet, so it cannot carry a label. **Immature cohorts were removed too**, which is the less obvious filter and the more important one: a loan only enters the study once its full contractual term has elapsed. A 36-month loan issued in mid-2017 that currently reads "Fully Paid" is a real observation, but its cohort is still developing and that cohort's eventual defaults are disproportionately still sitting in "Current" and "Late." Keeping the resolved ones and dropping the rest is a survivorship filter that biases measured loss *downward*. This is the same discipline as a reserving cut-off — you do not book an accident year at ultimate before it has developed.

That leaves **732,629 fully seasoned, resolved loans carrying \$9.67 billion of funded principal**, of which 14.9% charged off and 8.0% of principal was lost net of recoveries.

Then the leakage screen. The published file is a record of each loan's whole life, not a snapshot of underwriting, so information about the future is scattered through it. Thirty-eight columns were excluded in three named groups:

- **Payment performance** — how much was actually repaid, recoveries, last payment date. This is the answer, not a predictor.
- **Post-funding credit refresh** — the borrower's FICO score *today*, which for a defaulted borrower has usually already collapsed.
- **Distress flags** — hardship plans, settlement flags. These fields exist only *because* a loan went bad.

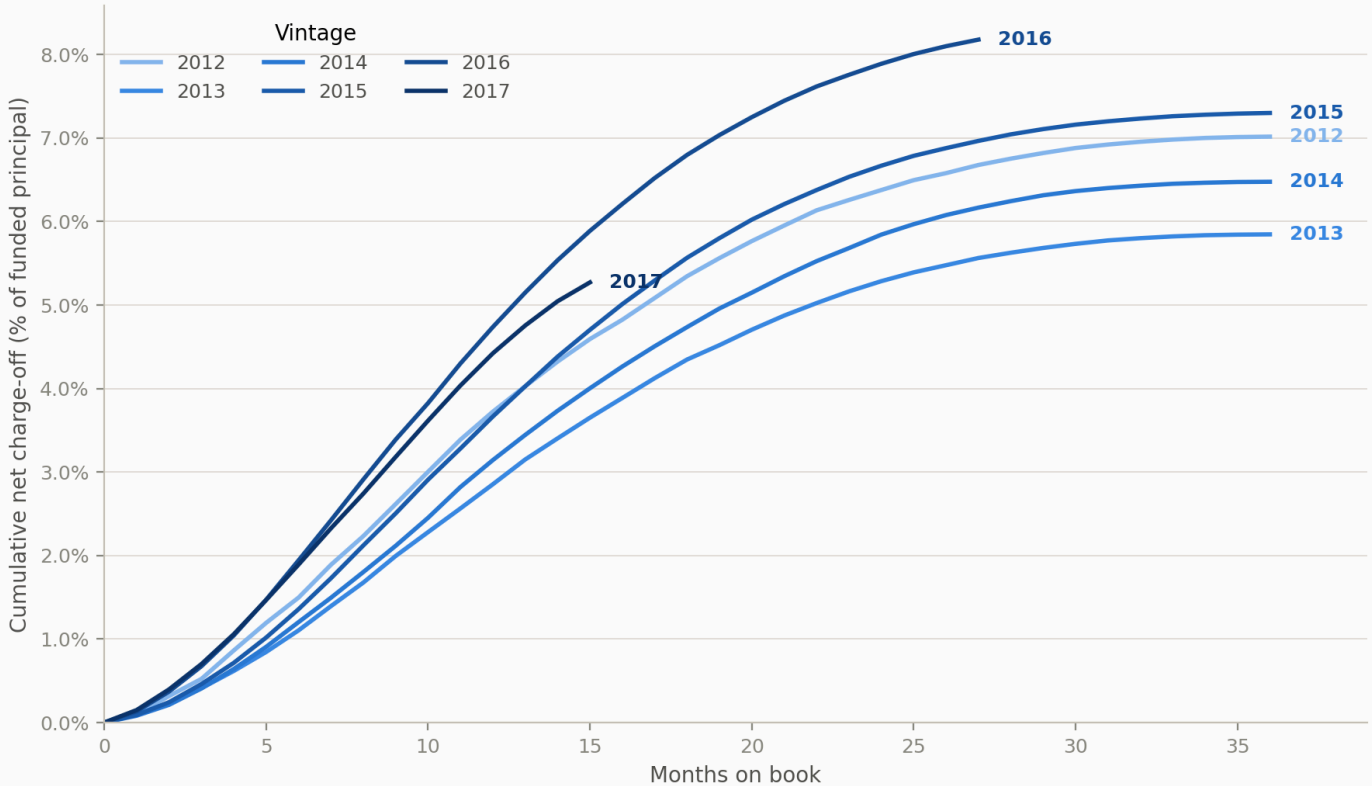
How badly does this matter? `debt_settlement_flag` is set for borrowers who negotiated a settlement, which happens after they stop paying. In the raw file it implies charge-off **100% of the time**. A model handed that column reports a near perfect AUC and has learned nothing about credit. Excluding these fields — and saying which and why — is the difference between a loss study and an accident.

One deliberate asymmetry: repayment and recovery amounts are *forbidden as predictors* and *required to measure the outcome*. A column can be both, and keeping that distinction straight is what lets the study talk in dollars rather than in classification labels.

2. Vintage curves: what the headline number hides

Group loans by the year they were originated, then track cumulative loss against **months on book** rather than calendar date. Every cohort is now compared at the same age, and the shape of underwriting change becomes visible.

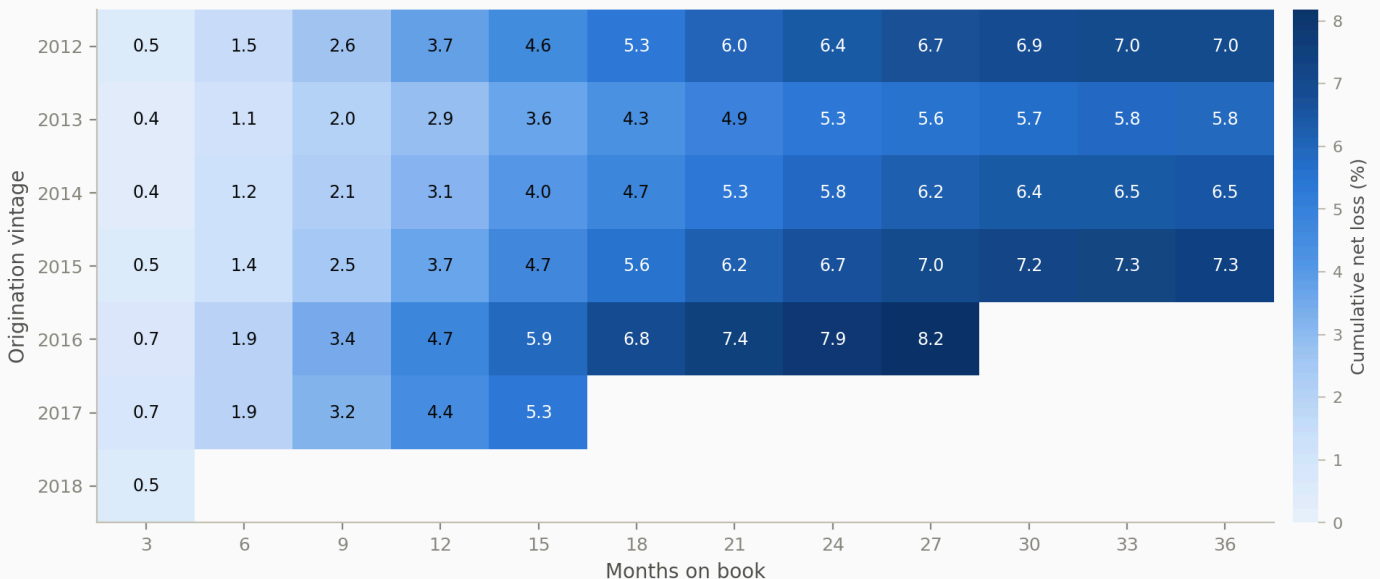
Vintage curves: cumulative net charge-off by months on book LendingClub 36-month loans, 2012-2017 origination cohorts



Each line is one origination cohort developing its losses over time — the credit analog of a paid-loss development triangle. Shorter lines are younger cohorts that have not yet been observed to 36 months.

Read as a triangle — cohorts down the side, development age across the top, blank lower-right where the future has not happened yet — this is structurally identical to a paid-loss development triangle.

Cumulative net charge-off by vintage and months on book 36-month loans · % of funded principal



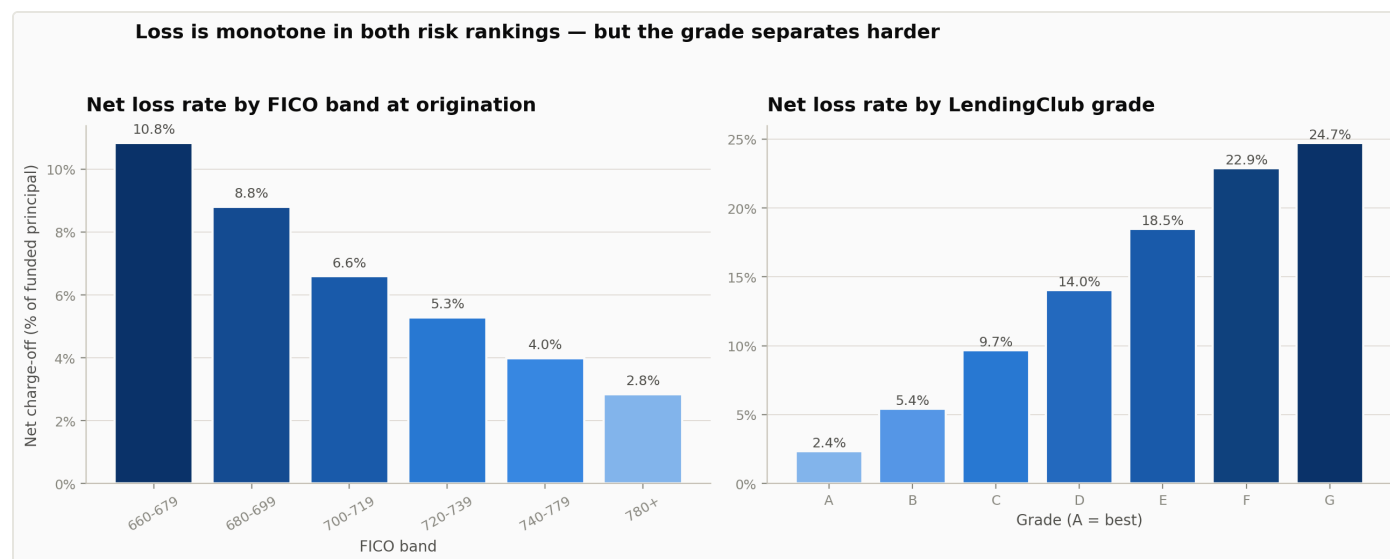
The findings:

- **2013 was the best book LendingClub wrote**, finishing at 5.9% cumulative net loss at 36 months and sitting lowest at every development age.
- **Credit loosened steadily from 2013 to 2016**. Loss at 12 months on book climbed 2.9% → 3.1% → 3.7% → 4.7% across the 2013–2016 vintages. Same product, same age, progressively more loss. Funded volume roughly doubled between 2013 and 2015 and doubled again by 2016; growth and deterioration arrived together, which is the usual pattern and the usual warning.
- **2016 is the worst cohort on the board**. At 24 months it had lost 7.9% of principal — more than the *fully developed* 36-month loss of the 2013 and 2014 vintages.

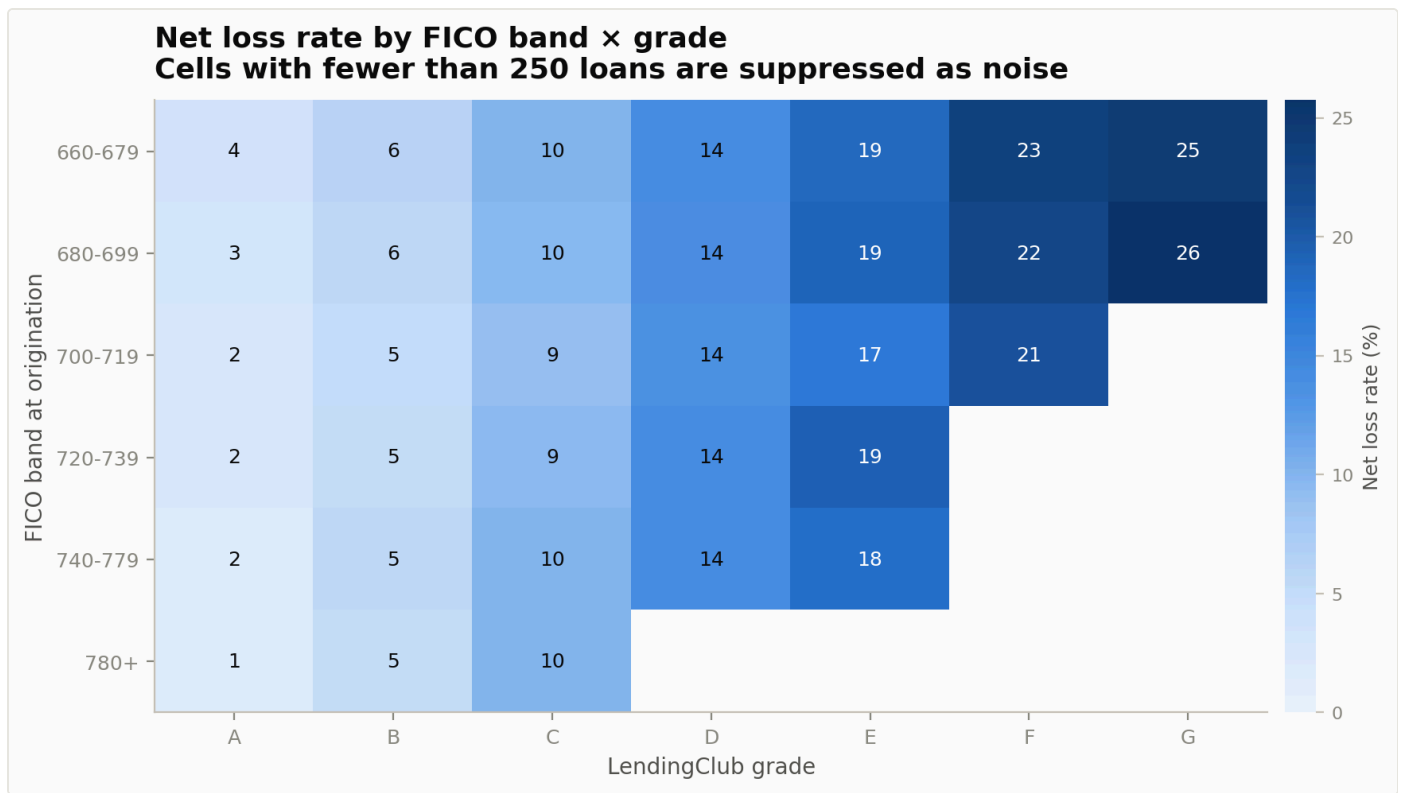
Why this matters more than a headline loss rate. The whole-portfolio net loss rate is 8.0%. That single number blends a 5.9% cohort with one tracking toward 9%, and it says nothing about direction. The triangle does. This is exactly why reserving is done by accident year rather than on the aggregate paid number: the average hides the trend, and the trend is the decision.

3. Where the loss sits — and the mix-versus-quality question

Splitting the book by FICO band and by LendingClub's own loan grade gives the credit version of a rate table.



Net loss runs from 2.8% in the 780+ FICO band to 10.8% in 660-679; from 2.4% in grade A to 24.7% in grade G. Both rankings are monotone, but **the grade separates far harder than the score** — and the two-way view explains why.

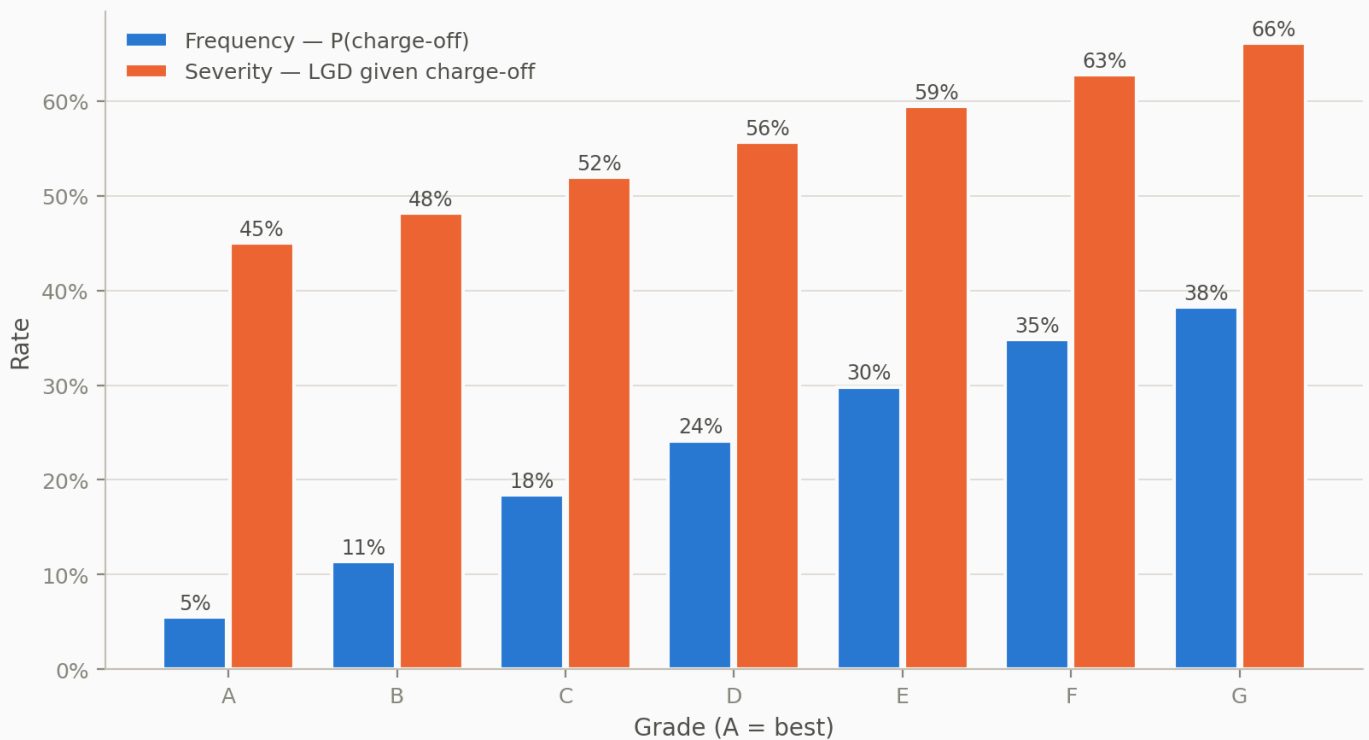


Read **across a row**: hold FICO fixed and loss still climbs steeply with grade — a 700–719 borrower graded A loses 2%, the same borrower graded E loses 17%. Read **down a column**: hold grade fixed and FICO does almost nothing. Grade C is 9–10% at every score band.

The grade is not a repackaged FICO score. It carries independent information — DTI, income verification, utilisation, inquiries — that no single bureau variable reproduces.

Frequency and severity are not the same size of problem.

Frequency swings 7× across the grade scale. Severity moves 1.5×. Which is why credit modeling is overwhelmingly a frequency problem



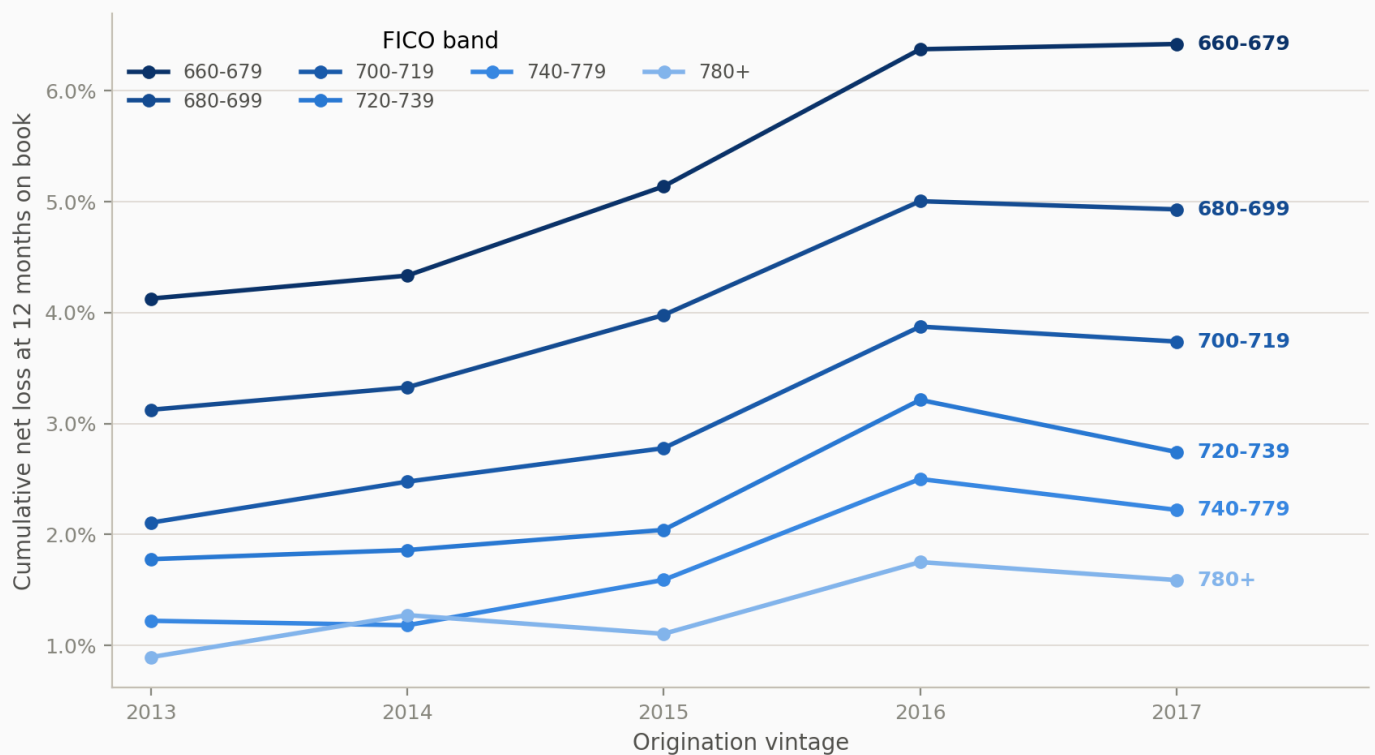
Across the grade scale, the probability of default swings **7×** (5.5% to 38%) while loss given default moves **1.5×** (45% to 66%). Both rise with risk, but only one of them is the variable term. That asymmetry is the structural fact behind everything that follows: unsecured consumer credit is a **frequency** problem, and modeling effort belongs there rather than in recovery and workout.

It is also a genuine difference from the catastrophe side. A property cat book is the mirror image — low, lumpy frequency and enormously variable severity, with the whole tail living in the severity term. The machinery transfers; the shape of the risk does not.

Does the price cover the loss? Converting the annual coupon and cumulative loss onto a common footing, gross spread stays positive at every grade and *widens* as credit worsens — the shape of a rate table doing its job. Two caveats belong with that: gross spread still has to pay servicing, funding and the cost of capital, and the wider spread at the bottom comes with far more loss volatility. Whether that is a good trade is a capital question, which is section 5.

The mix question. Every FICO band deteriorated into 2016 — and the *prime* bands deteriorated proportionally more (740–779 doubled; 660–679 rose 55%). That rules out a mix shift as the explanation: the credit quality of a 740-FICO borrower genuinely got worse at the same score.

The 2016 deterioration was broad, not concentrated Loss at 12 months on book by FICO band, 36-month loans



The 2017 vintage then looked better in aggregate — 4.42% at 12 months versus 4.74%.

Decomposing that 0.32-point improvement the way a loss-ratio movement gets split into rate and mix:

Effect	Contribution
Mix — reweighting toward better FICO bands	−0.19 pts (60%)
Within-band — genuinely tighter standards	−0.13 pts (40%)

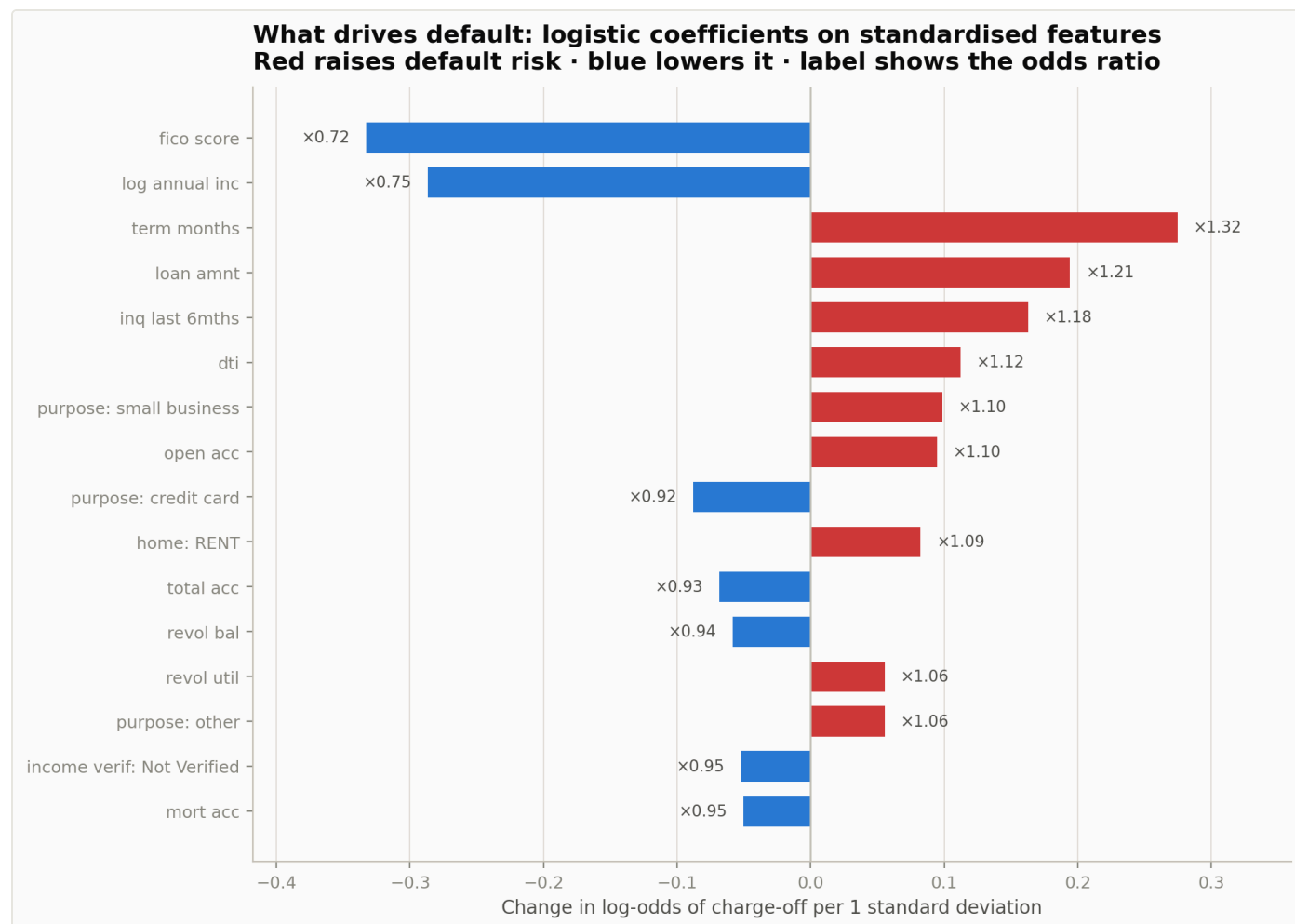
Most of the 2017 improvement was a volume decision, not an underwriting one.

LendingClub shifted the book toward better credit — the 660–679 band fell from 32% to 29% of principal while the 740+ bands grew from 11% to 15%. Within-band standards improved only modestly, and at the bottom of the score range 2017 was marginally *worse* than 2016. A mix shift reverses with the next growth target; tighter underwriting persists. Reading the aggregate alone, you would have called this a credit turnaround.

4. The default model — and what interpretability actually costs

A logistic regression on origination-time borrower features, validated **out of time**: trained on vintages through 2014, tested on 2015–2016. Not a random split — the job is to underwrite loans

that have not been written yet, and the test cohorts are ones we already know were worse.



Because the inputs are standardised, each coefficient reads as "what happens to default odds if this feature moves one standard deviation." Six effects dominate, and every one is something a human underwriter would recognise:

- **FICO is the strongest single feature** — one standard deviation (~32 points of score) cuts default odds by 28%.
- **Income lowers risk almost as much as FICO does.** Capacity to pay is nearly as informative as credit history.
- **Term is the strongest risk-increasing feature.** Sixty months rather than thirty-six raises default odds 32% per standard deviation — part genuine duration risk, part selection, since borrowers who need the smaller payment are more stretched to begin with.
- **Loan size, recent credit inquiries and DTI all raise risk**, in that order.
- **Stated purpose carries real signal.** Relative to debt consolidation, small-business borrowing is riskier and credit-card refinancing is safer — a borrower consolidating card debt at a lower rate is doing something economically sensible; someone funding a business with unsecured personal credit is taking business risk onto a consumer balance sheet.

Nothing has the wrong sign. Nothing needs a footnote. That property — every coefficient defensible out loud, without a tool — is what is being bought, and it is what a model-risk function and a fair-lending adverse-action requirement both demand.

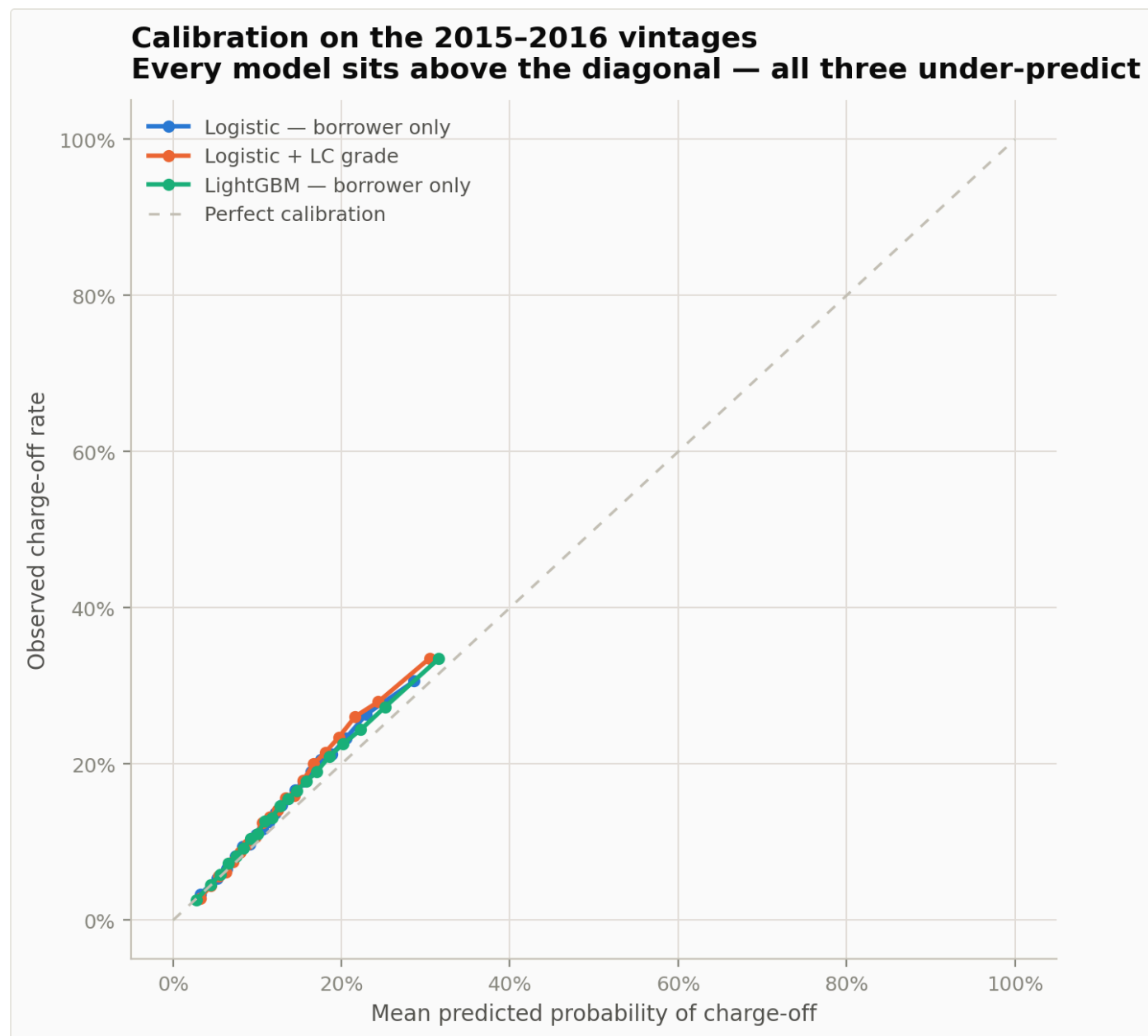
The benchmark

Model	Out-of-time AUC	KS
Logistic — borrower features	0.657	0.226
LightGBM — same features	0.675	0.252
Logistic + LendingClub grade	0.684	0.270

Gradient boosting beats the logistic model by 0.018 AUC. That is a real gap, not a rounding error — but it is small in the terms that matter, since it moves few loans across any threshold a lender would actually set, and it costs readable coefficients, direct reason codes and a straightforward model-risk review.

The sharper result is the third row. Adding the lender's own grade to the *logistic* model beats the boosted model on borrower features alone. Better information beat a better functional form — the usual answer in credit, and the reason underwriting teams spend their budget on data rather than on architecture.

The most instructive result: the model under-predicts



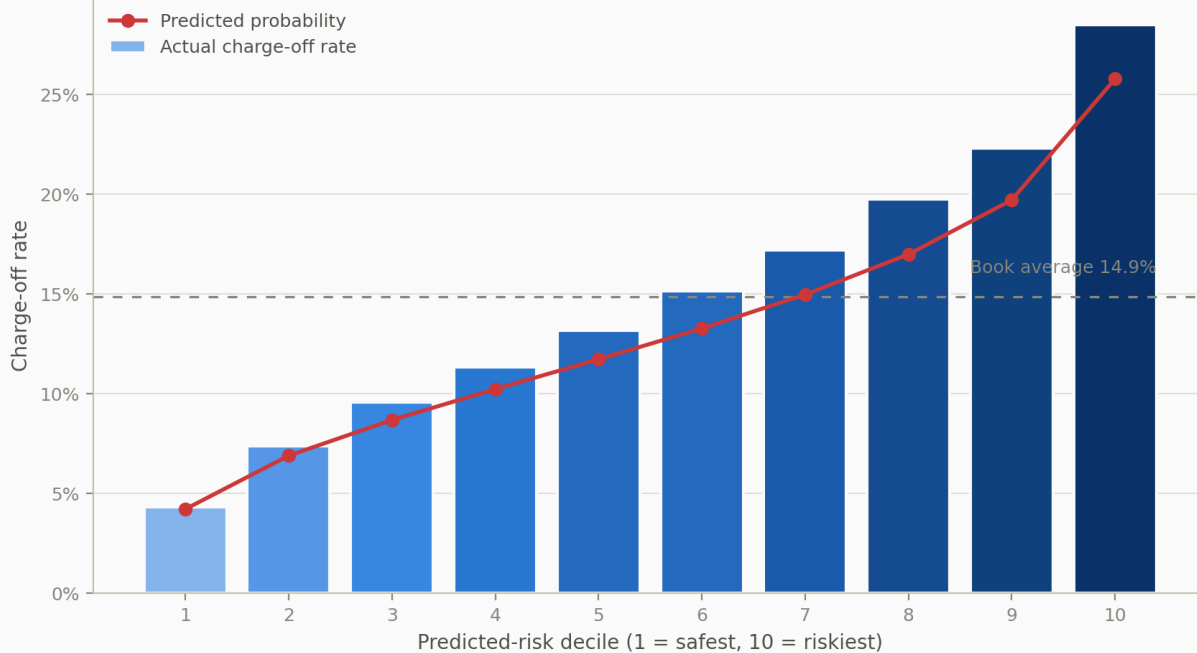
On the out-of-time test the model predicts a 13.3% default rate against 14.9% actual — **12% too low in relative terms**.

That is not a bug; it is section 2 showing up in the model. The 2015-2016 vintages were genuinely worse than the 2007-2014 book at the same observable characteristics, and nothing in the feature set could have said so.

This is why credit models in production are **recalibrated frequently and refit rarely**. Rank-ordering is stable and transfers out of time — the decile lift table is cleanly monotone with no inversions on cohorts the model never saw. The *level* drifts with the credit cycle and has to be reset against recent performance. The reinsurance analogy is exact: the relativities hold, the base rate has moved and needs trending.

Decile lift on the out-of-time test set

Rank-ordering is clean and monotone; the predicted line sits below actuals throughout



5. The portfolio view: expected loss, and where the tail comes from

Which book, and why not all of it. The portfolio simulated here is the **2015-2016 holdout** — 333,721 loans, \$4.32 billion — not the full 732,629 from section 1. The loan-level probabilities have to come from a model that never saw these loans: a loss distribution built on in-sample fitted probabilities is too narrow, because the model has already been shown the outcomes it is being asked to predict. The counts reconcile exactly — 398,906 training loans plus 333,721 holdout loans is the full 732,627 that survives the sub-660 FICO exclusion.

Aggregating those loan-level probabilities using the standard three-term product (probability of default × loss given default × exposure), and recalibrating them as section 4 argued:

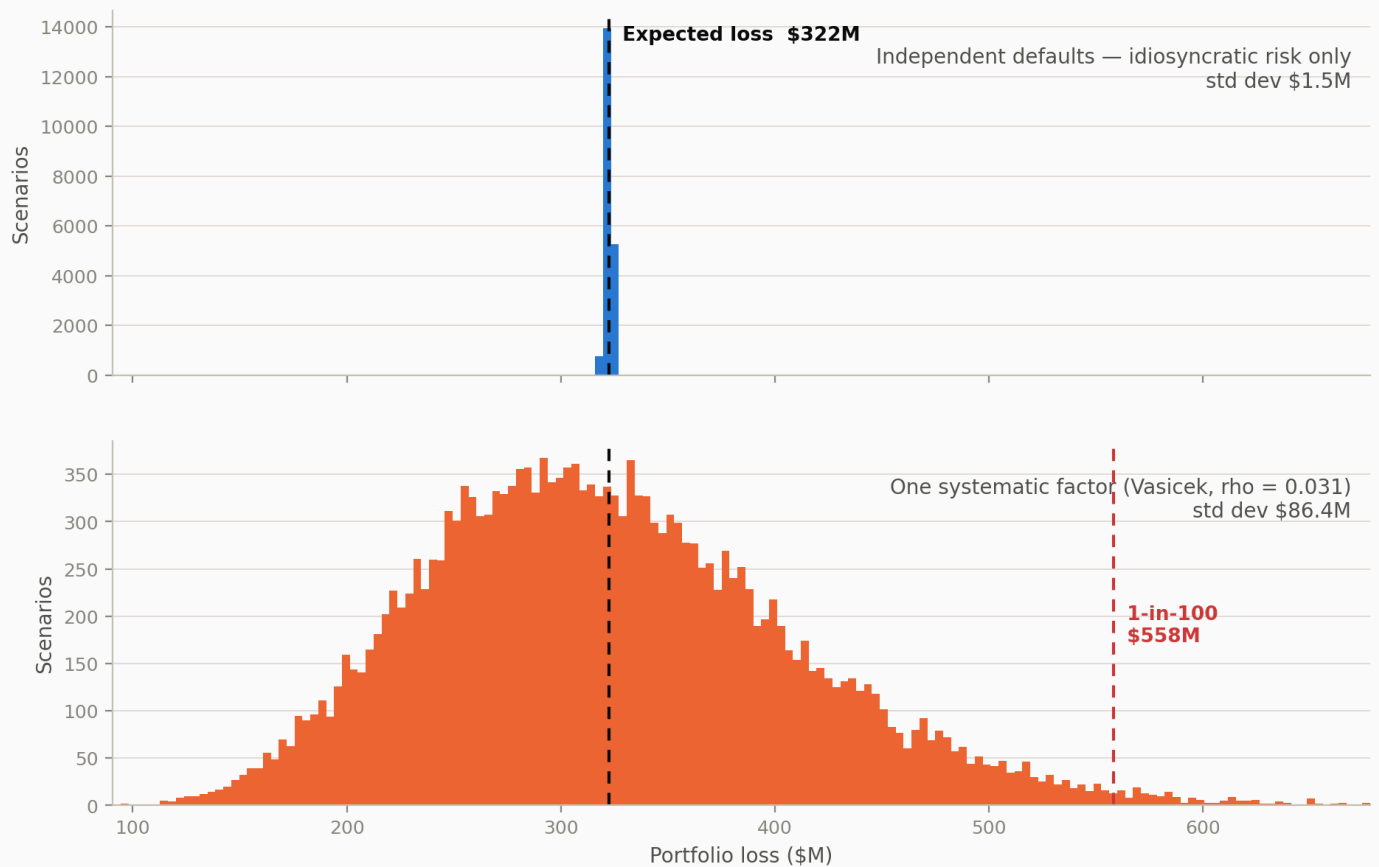
Modeled expected loss: \$322.4M (7.46% of exposure). Actual realised net loss: \$322.0M (7.45%).

That is the AAL. It is also the easy part. A point estimate tells you what to charge; it says nothing about what to hold capital against.

The tail is made of correlation

Two simulations of the same book, 20,000 scenarios each, with identical expected loss:

The tail is made of correlation, not of credit risk
20,000 simulated outcomes · same \$4.3B book · same expected loss · same x-axis

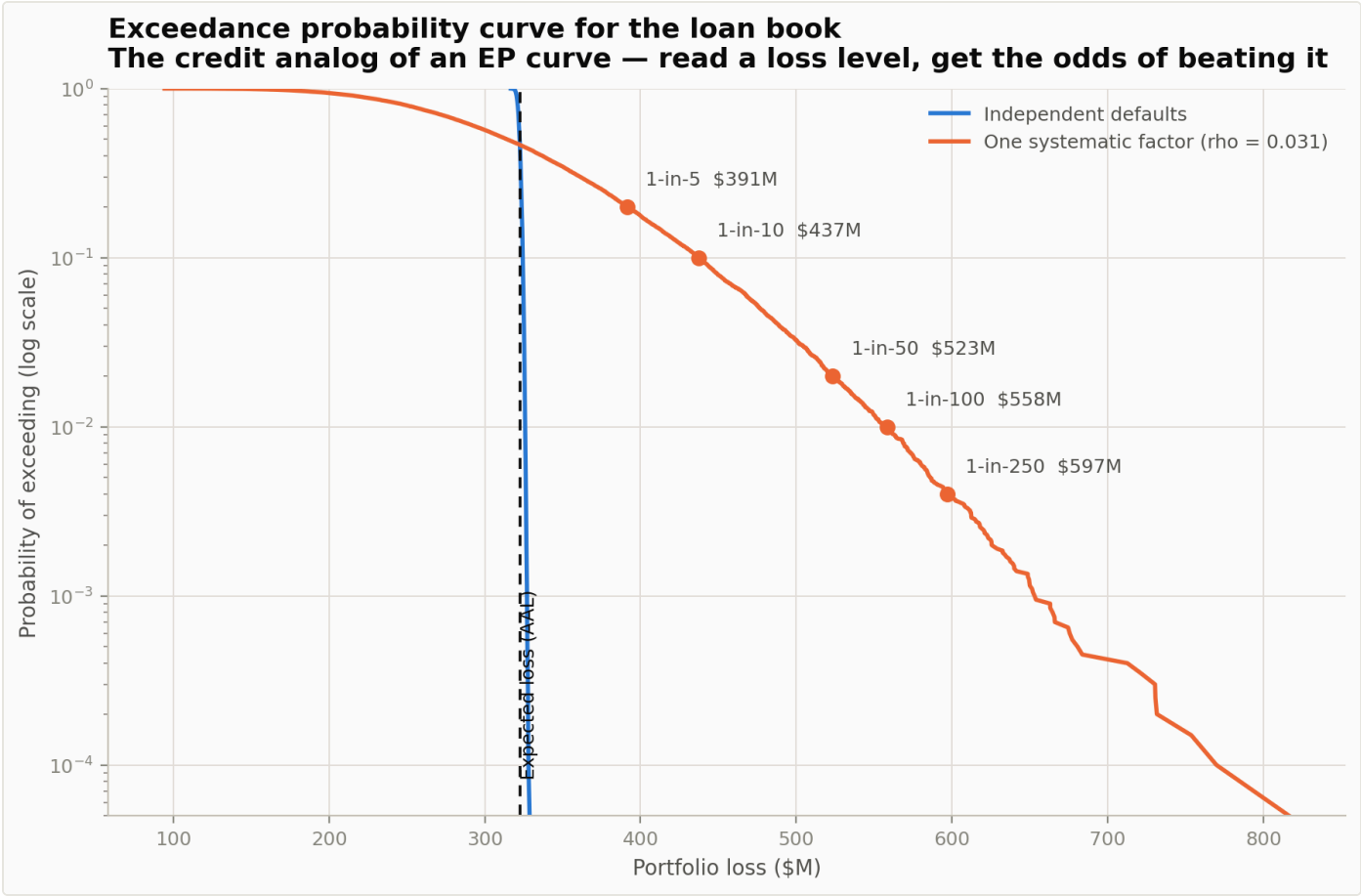


Independent defaults — every loan flips its own coin — produce a spike. Standard deviation \$1.5M on a \$322M mean. With a book this large the law of large numbers does its job almost perfectly, and a lender would need essentially no capital.

One systematic factor — the Vasicek model underneath the Basel capital formula, where every borrower's creditworthiness has a common economic component — produces a standard deviation of \$86.4M and a long right tail.

That factor is the event. It is the credit-portfolio equivalent of a hurricane damaging ten thousand houses on the same afternoon. **Diversification does not protect against a common cause**, which is why writing more Florida wind does not diversify a Florida wind portfolio, and why the 2008 mortgage models failed the way they did.

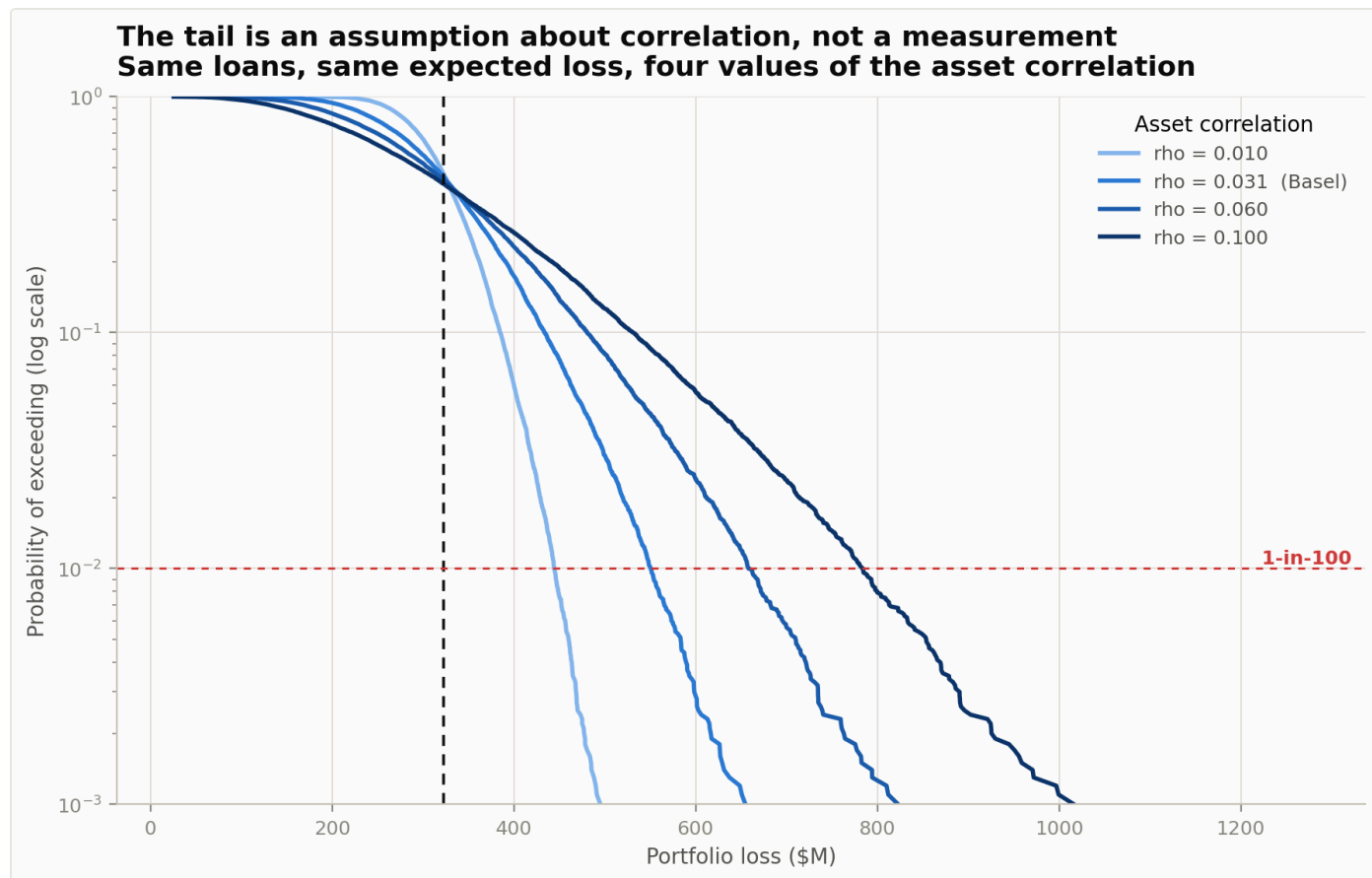
The EP curve



Return period	Portfolio loss	Loss rate	× expected loss
1-in-5	\$391M	9.1%	1.21×
1-in-10	\$437M	10.1%	1.36×
1-in-50	\$523M	12.1%	1.62×
1-in-100	\$558M	12.9%	1.73×
1-in-250	\$597M	13.8%	1.85×

Expected loss goes into the price. Everything above it, out to a chosen return period, is **unexpected loss** — what equity is held against, and the number that decides whether the spread from section 3 is an adequate return. Two treaties with the same expected loss and different tails are not the same trade, and neither are two loan books.

And the tail is an assumption, so it is shown as one



Moving the asset correlation from 0.01 to 0.10 lifts the 1-in-100 from 1.4× expected loss to 2.4× — a 76% increase in the capital number — while **expected loss does not move at all**. A pricing model and a capital model can agree completely on the mean and disagree by a factor of two on the thing that matters.

Checked against reality, the model is about three times more dispersed than the four observed vintage outcomes (27% coefficient of variation versus 10% realised). Both sides of that gap are informative: the Basel correlation is a *deliberately conservative* regulatory parameter, and four cohorts from 2012–2015 is a benign, tiny sample — **the window does not contain 2008**. Neither figure is a measurement of the tail. The realised dispersion is a floor built from a calm period; the regulatory figure is a prudent ceiling; the honest way to narrow the gap is to stress the book against a downturn rather than trust either number. A cat modeler with a short historical event set faces exactly this problem and does exactly this.

6. What I'd do next

Four things this study does not do, in the order I'd tackle them:

1. **Stress the tail against a downturn.** The correlation is calibrated on a window containing no recession, so the 1-in-100 is an extrapolation from calm weather. Overlaying the 2007–2009 experience — or a macro scenario — would replace it with something defensible.
2. **Model loss given default rather than fixing it at 52%.** Severity is stable enough to hold constant across the book, which is what justifies the shortcut here, but it runs 45% in grade A to 66% in grade G. A grade- and term-conditional LGD would sharpen expected loss exactly where the book is worst, and it is the step that moves this from a frequency model to a full expected-loss model.
3. **Make the default probability macro-conditional.** The model under-predicts out of time by 12%, and section 4 patches that with a single recalibration scalar. A time-varying intercept, or unemployment as a cohort-level covariate, would explain the drift structurally instead of absorbing it.
4. **Build the 60-month book its own triangle.** It is held out of the vintage curves deliberately, to keep loss timing comparable — but it is the structurally riskier half of the portfolio and it deserves the analysis rather than the exclusion.

The first two matter most. Everything in section 5 above the mean is an assumption about correlation, and everything below it assumes severity is a constant.

What this demonstrates

Real loan-level data, cleaned with an explicit leakage screen; loss measured in dollars and developed by cohort; a default model whose every coefficient can be defended out loud, honestly validated out of time and honestly reported as miscalibrated; and a portfolio loss distribution whose tail is traced to its actual source and stress-tested against the assumption driving it.

The techniques are the ones a credit risk team uses daily. They are also, under different names, the ones a catastrophe and ILS team uses daily — vintage curves are development triangles, expected loss is AAL, the loss distribution is an EP curve, and the tail in both comes from correlation rather than from the risk of any single exposure.